KNOWLEDGE INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

Approved by AICTE, Affiliated to Anna University,

Accredited by NAAC and NBA (B.E: Mech., ECE, EEE & CSE)

Kakapalayam (PO), Salem – 637 504

Department of Electronics and Communication Engineering

LABORATORY MANUAL

Course Title	**: MEMS Design**
Course Code	**: CEC340**
Regulation	**: 2021**

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COLLEGE VISION

To be a world class institution to impart value and need based professional education to the aspiring youth and carving them into disciplined world class professionals who have quest for excellence, achievement orientation and social responsibilities.

COLLEGE MISSION

To promote academic growth by offering state-of-the-art undergraduate, postgraduate and doctoral programs and to generate new knowledge by engaging in cutting-edge research.

VISION OF THE DEPARTMENT

- To produce competent Electronics and Communication Engineers by imparting quality education to meet the industry requirements and for serving the societal needs

MISSION OF THE DEPARTMENT

- To develop appropriate facilities for promoting research activities
- To inculcate leadership qualities among students for self and societal growth
- To nurture students on emerging technologies for serving industry needs through industry institute interface
- To enrich teaching learning process by transforming young minds to be resourceful engineers

PROGRAMME EDUCATIONAL OBJECTIVES (PEOS)

The Program Educational Objectives(PEOs) for the Electronics and Communication Engineering program describe accomplishments that graduates are expected to attain within five to seven years after graduation

PEO – 1: To enable graduates to pursue research, or have a successful career in academia or industries associated with Electronics and Communication Engineering, or as entrepreneurs

PEO – 2: To provide students with strong foundational concepts and also advanced techniques and tools in order to enable them to build solutions or systems of varying complexity

PEO – 3: To prepare students to critically analyze existing literature in an area of specialization and ethically develop innovative and research oriented methodologies to solve the problems identified.

PROGRAMME SPECIFIC OUTCOME (PSOS)

After the successful completion of B.E programme in Electronic and Communication Engineering, the graduates will be able to

PSO – 1: Use signal processing concepts and tools to provide solutions to real time problems.

PSO – 2: Use embedded system concepts for developing IoT applications

PSO – 3: Use the concepts of analog and digital electronics to design and implement VLSI circuits.

PROGRAMME OUTCOMES (POS)

The graduates of Electronics and Communication Engineering will be able to:

PO – 1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO – 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO – 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO – 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO – 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO – 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO – 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO – 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO – 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams and in multidisciplinary settings.

PO – 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation make effective presentations, and give and receive clear instructions.

PO – 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO – 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Date: Expt. No. : 1	**Design and simulation of piezoelectric cantilever**

AIM:

To Design and simulation of a piezoelectric cantilever using a beam configuration

SOFTWARE REQUIRED:

- MATLAB
- Windows 11 OS

THEORY:

A piezoelectric cantilever beam converts mechanical vibrations into electrical signals using the piezoelectric effect. It is widely used in energy harvesting and sensing applications. The goal of this experiment is to simulate the dynamic behavior of such a beam and estimate its voltage output based on mechanical strain.

The piezoelectric cantilever operates on two key physical principles:

Beam Vibration Mechanics:

When subjected to vibrations or dynamic loading (like a tip force or base motion), the beam bends.

The bending induces strain along the beam length, particularly at the fixed end.

Piezoelectric Effect:

The piezoelectric layer (bonded to the beam) undergoes mechanical strain.

Due to the direct piezoelectric effect, this strain generates electric charge on the electrodes attached to the piezo material.

This charge creates a measurable voltage, which can be used for:

Sensing vibration,

Harvesting energy, or

Actuating motion (reverse effect).

The voltage output from the piezoelectric material is a key parameter to be estimated based on the mechanical strain applied to the cantilever beam.

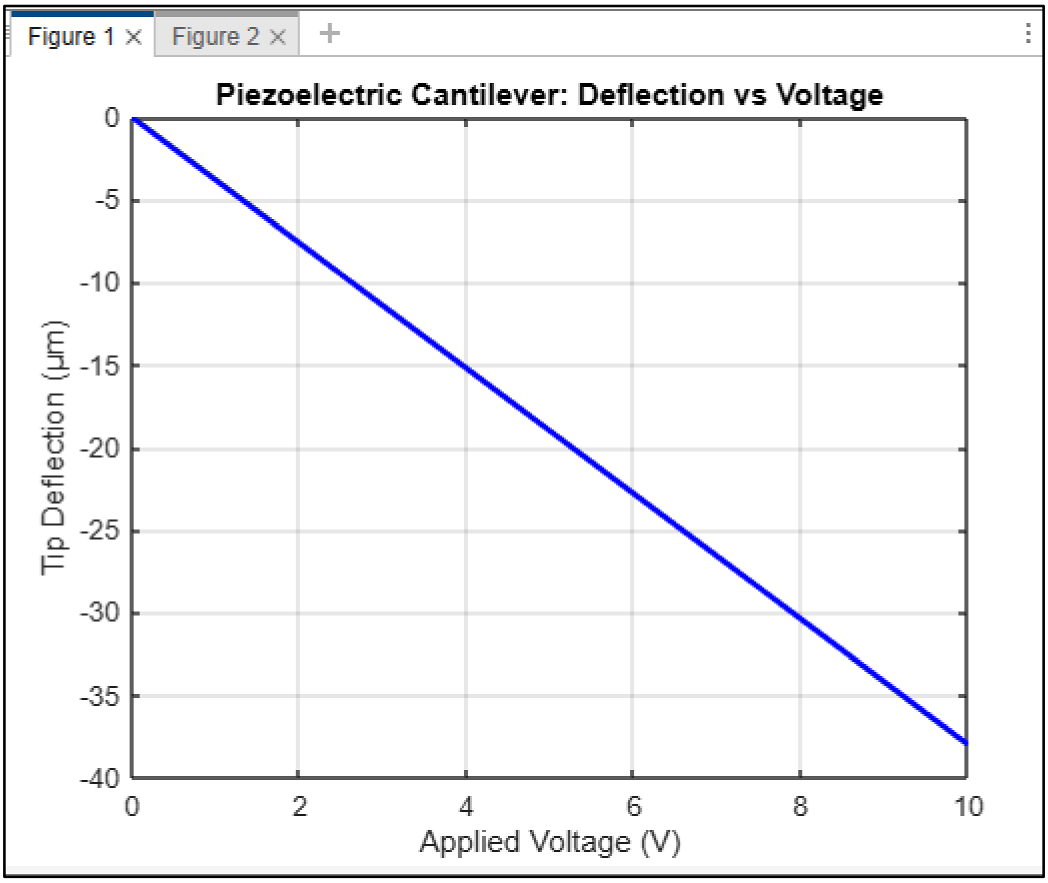
PROGRAM:

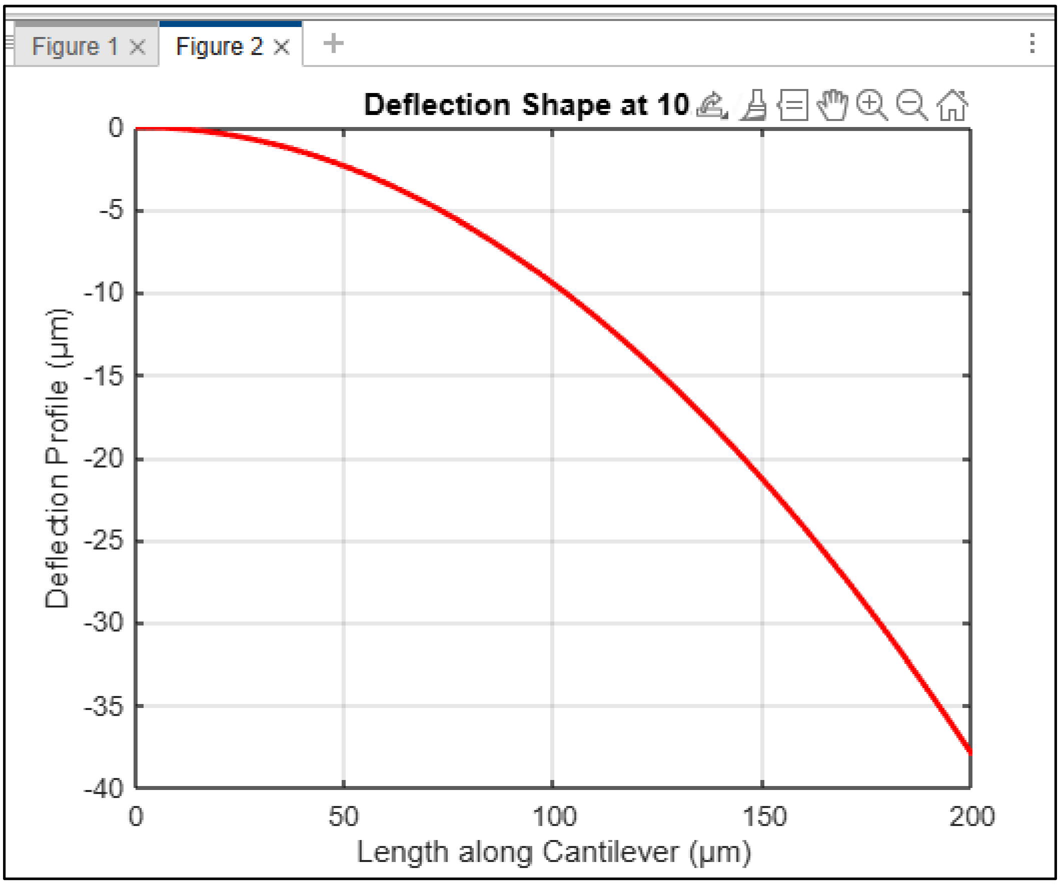
```
% Piezoelectric Cantilever Design and Simulation

clc; clear;

% === 1. Geometry and Material Parameters ===

L = 200e-6;       % Length (m)

W = 50e-6;        % Width (m)

H_sub = 2e-6;     % Substrate thickness (Silicon) (m)

H_piezo = 1e-6;   % Piezoelectric layer thickness (PZT) (m)

H_total = H_sub + H_piezo;


% Material properties

E_sub = 130e9;    % Young's modulus of Silicon (Pa)

E_piezo = 66e9;   % Young's modulus of PZT (Pa)

rho = 7800;       % Density (kg/m^3)

% Piezoelectric properties

d31 = -190e-12;          % Piezoelectric coefficient (m/V)

epsilon_r = 1000;         % Relative permittivity

epsilon_0 = 8.854e-12;    % Vacuum permittivity


% === 2. Mechanical Properties ===

% Equivalent Young's modulus (simplified mixture)

E_eq = (E_sub * H_sub + E_piezo * H_piezo) / H_total;

% Moment of inertia for rectangular cross-section

I = (W * H_total^3) / 12;

% Effective mass of cantilever

m_eff = 0.24 * rho * W * H_total * L;

% Resonant frequency (1st mode)
```

```
f_res = (1 / (2 * pi)) * sqrt((3 * E_eq * I) / (m_eff * L^3));
fprintf('--- Mechanical Properties ---\n');
fprintf('Resonant Frequency: %.2f kHz\n', f_res / 1e3);

% === 3. Sensor Mode: Voltage Output from Vibration ===
z_tip = 1e-6;  % Tip displacement (m)
strain = (6 * z_tip * (H_total / 2)) / L^2;
E_field = strain / d31;         % Electric field (V/m)
V_out = E_field * H_piezo;      % Output voltage (V)
fprintf('\n--- Sensor Mode ---\n');
fprintf('Tip Displacement: %.2f µm\n', z_tip * 1e6);
fprintf('Generated Voltage: %.2f mV\n', V_out * 1e3);

% === 4. Actuator Mode: Deflection vs Applied Voltage ===
V_in = 5;  % Input voltage (V)
strain_piezo = d31 * V_in / H_piezo;
z_actuation = (3 * strain_piezo * L^2) / (2 * H_total);

fprintf('\n--- Actuator Mode ---\n');
fprintf('Applied Voltage: %.2f V\n', V_in);
fprintf('Tip Deflection: %.2f µm\n', z_actuation * 1e6);

% === 5. Sweep Voltage and Plot Deflection ===
V_range = 0:0.5:10;
deflections = (3 * d31 * V_range / H_piezo) * (L^2) / (2 * H_total);
figure;
plot(V_range, deflections * 1e6, 'b-', 'LineWidth', 2);
```

```
xlabel('Applied Voltage (V)');
ylabel('Tip Deflection (µm)');
title('Piezoelectric Cantilever: Deflection vs Voltage');
grid on;

% === 6. Plot Cantilever Profile at Max Voltage ===
x = linspace(0, L, 100);
z_profile = deflections(end) * (x / L).^2;  % Approx. parabolic shape
figure;
plot(x * 1e6, z_profile * 1e6, 'r-', 'LineWidth', 2);
xlabel('Length along Cantilever (µm)');
ylabel('Deflection Profile (µm)');
title(sprintf('Deflection Shape at %.1f V', V_range(end)));
grid on;
```

OUTPUT WAVEFORM:





Analysis of a piezoelectric sensor based on a cantilever beam configuration

```
% Beam Geometry & Material Properties

L = 200e-6;         % Length (m)

b = 50e-6;          % Width (m)

h = 2e-6;           % Thickness (m)

E = 160e9;          % Young's Modulus (Pa)

d31 = -190e-12;     % Piezoelectric strain coefficient (m/V)

eps33 = 15.93e-9;   % Permittivity (F/m), typical for PZT

t_piezo = 1e-6;     % Thickness of piezo layer (m)

% Loading

F = 1e-6;           % Point load at free end (N)

% Calculations

I = (b * h^3) / 12;                     % Moment of inertia

delta = (F * L^3) / (3 * E * I);        % Tip deflection (m)

strain_max = (6 * F * L) / (E * b * h^2);   % Max strain at fixed end

% Electric field and voltage generated

% Voltage V = strain * t_piezo / d31 (simplified)

V_out = abs(strain_max) * t_piezo / abs(d31);

% Display results

fprintf('Tip deflection: %.3f µm\n', delta * 1e6);

fprintf('Maximum strain: %.3f microstrain\n', strain_max * 1e6);

fprintf('Estimated output voltage: %.3f V\n', V_out);
```

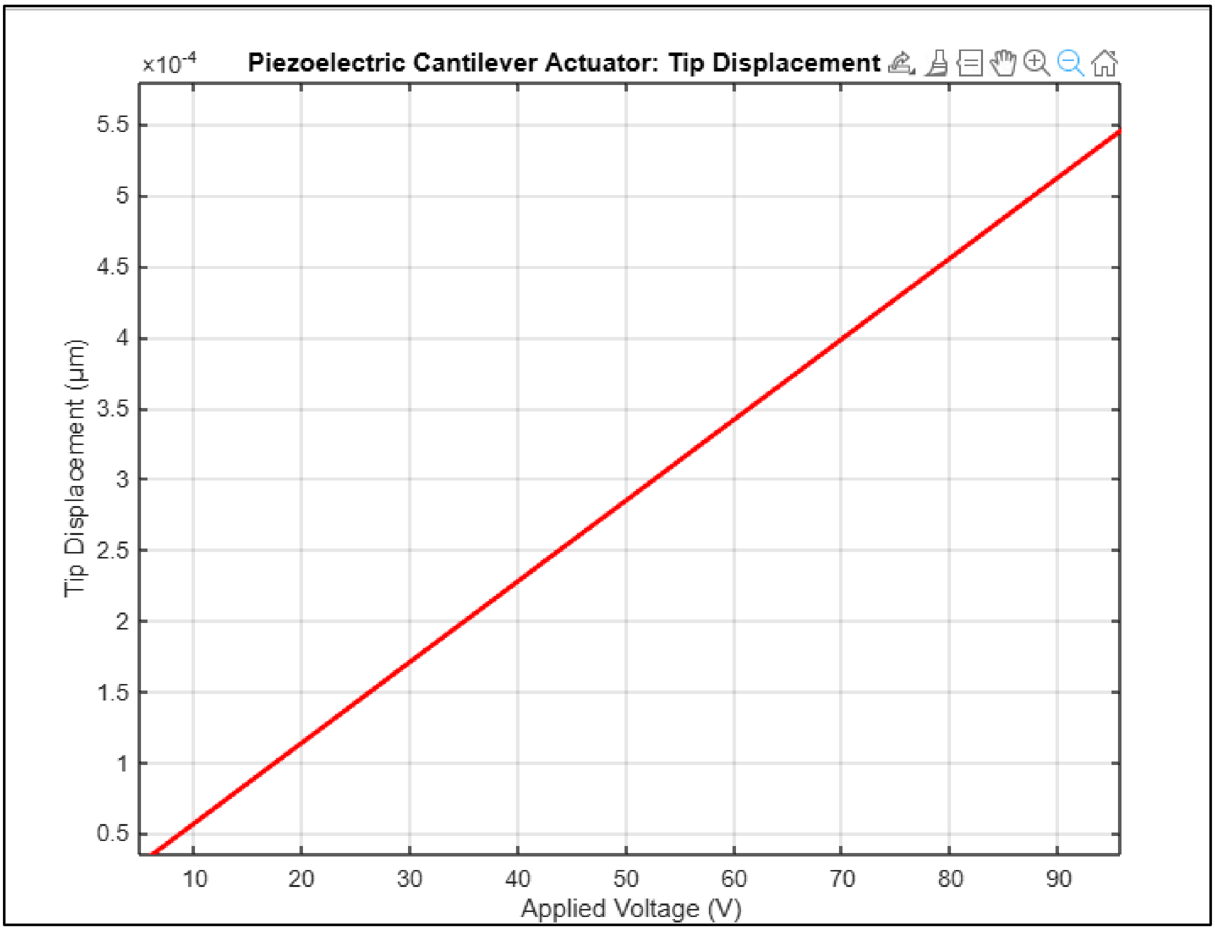
OUTPUT :

```
Command Window
>> Memsexp1_1
Tip deflection: 0.500 µm
Maximum strain: 37.500 microstrain
Estimated output voltage: 0.197 V
>>
```

Analysis of piezoelectric actuator based on a cantilever beam configuration

```
% --- Geometric and Material Properties ---

L = 200e-6;         % Beam length (m)

b = 50e-6;          % Beam width (m)

h_sub = 2e-6;       % Substrate thickness (m)

h_piezo = 1e-6;     % Piezo layer thickness (m)

E_sub = 160e9;      % Young's modulus of substrate (Pa)

d31 = -190e-12;     % Piezoelectric strain coefficient (m/V)


% --- Electrical Excitation ---

V = 0:5:100;        % Voltage range (V)


% --- Bending Induced Displacement (Analytical Approximation) ---

% Tip deflection due to piezo strain: Δ = (3 * d31 * V * L^2) / (2 * h_sub)

delta_tip = (3 * abs(d31) * V * L^2) ./ (2 * h_sub);

% --- Plotting Results ---

figure;

plot(V, delta_tip * 1e6, 'r-', 'LineWidth', 2);  % Convert to µm

xlabel('Applied Voltage (V)');

ylabel('Tip Displacement (µm)');

title('Piezoelectric Cantilever Actuator: Tip Displacement vs Voltage');

grid on;
```

OUTPUT WAVEFORM:



MATLAB WINDOWS:

- Files panel — Access your files.
- Workspace panel — Explore data that you create or import from files.
- Command Window — Enter commands at the command line, indicated by the prompt (>>).
- Sidebars — Access tools docked in the desktop and additional panels.

PROCEDURE:

1. Define Geometry and Material Properties.
2. Calculate Structural Parameters.
3. Define External Excitation Function.
4. Define the System Dynamics (ODE Function).
5. Calculate Generated Voltage.

INFERENCE:

RESULT:

Date: Expt. No. : 2	**Design and simulation of thermocouples**

AIM:

To design and simulation of thermocouples for various materials.

SOFTWARE REQUIRED:

- MATLAB
- Windows 11 OS

THEORY:

1. Common thermocouple types differ by metal combinations and temperature ranges:

Type	Metals	Temp Range	Seebeck Coefficient
Type K	Chromel – Alumel	-200 to 1350°C	~41 µV/°C
Type J	Iron – Constantan	-40 to 750°C	~50 µV/°C
Type T	Copper – Constantan	-200 to 370°C	~43 µV/°C
Type R/S	Platinum – Rhodium	0 to 1600°C	Low sensitivity (~10 µV/°C)

Selection Criteria:

Temperature range

Accuracy required

Environmental conditions (corrosive, oxidizing, etc.)

Cost

2. Junction Configuration

Grounded Junction: Fast response, good for liquids/gases, but possible electrical noise.

Ungrounded Junction: Electrically isolated; slower response.

Exposed Junction: Fastest response; suitable for low-corrosion environments.

3. Wire Gauge and Length

Thinner wires give faster response but are more fragile.

Longer wires introduce more resistance and signal degradation.

4. Sheathing and Protection

Common materials:

Stainless steel (for general use)

Inconel (high-temp environments)

Ceramic (extreme temperatures)

5. Cold Junction Compensation (CJC)

Modern thermocouple systems include cold-junction compensation, usually via an IC temperature sensor or reference table, because the thermocouple only measures a difference in temperature.

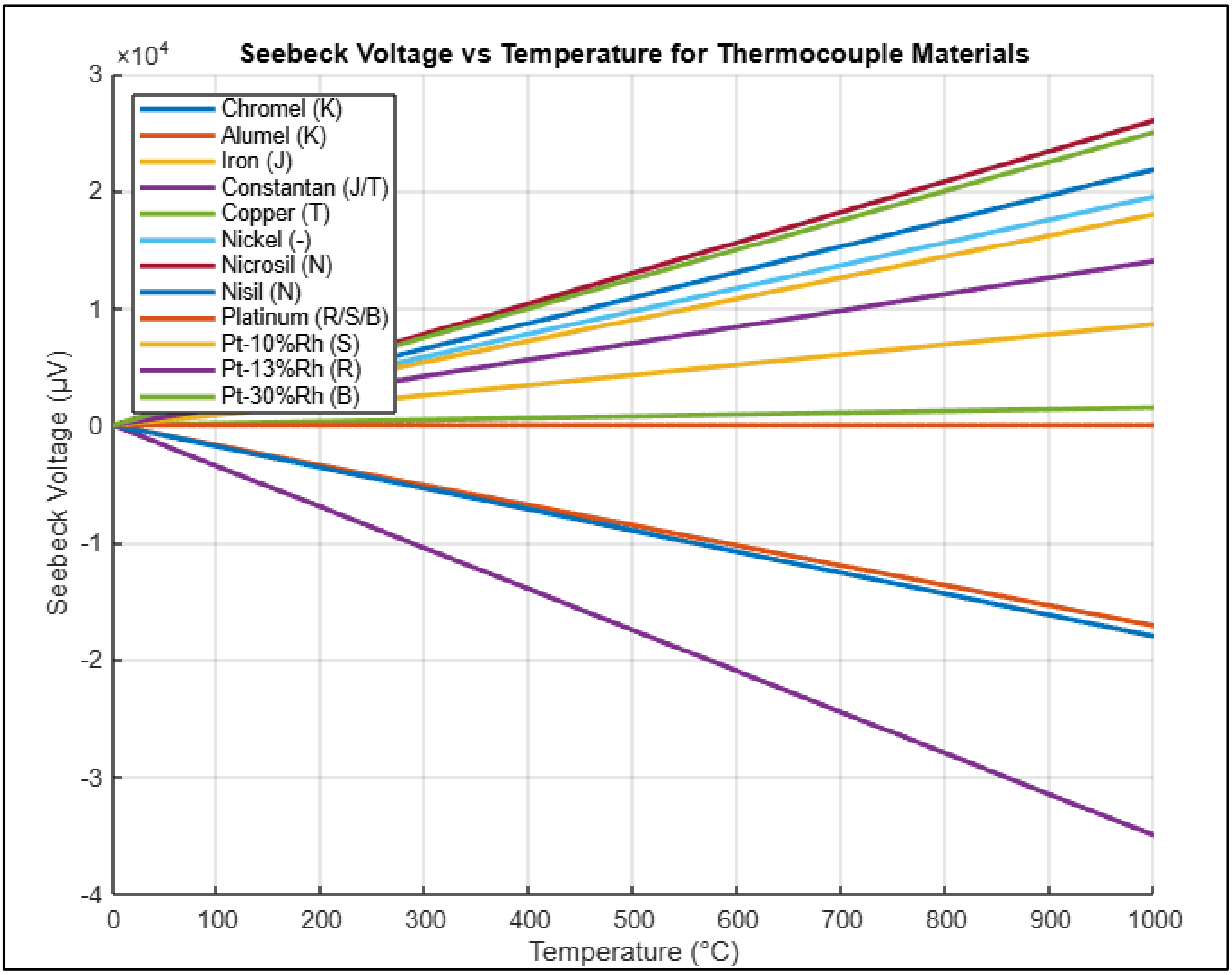
PROGRAM:

```
clc; clear;

% === Define Temperature Range (in °C) ===

T = 0:10:1000;  % Temperature vector (1 x M)

numTemps = length(T);

% === Define Thermocouple Material Properties ===

materials = {

    'Chromel', 'K', 21.8;

    'Alumel', 'K', -17.1;

    'Iron', 'J', 18.0;

    'Constantan', 'J/T', -35.0;

    'Copper', 'T', 1.5;

    'Nickel', '-', 19.5;

    'Nicrosil', 'N', 26.0;

    'Nisil', 'N', -18.0;
```

```
    'Platinum', 'R/S/B', 0.0;
    'Pt-10%Rh', 'S', 8.6;
    'Pt-13%Rh', 'R', 14.0;
    'Pt-30%Rh', 'B', 25.0;
};
% === Extract Data into Arrays ===
numMaterials = size(materials, 1);
materialNames = materials(:, 1);
materialTypes = materials(:, 2);
S_uV = cell2mat(materials(:, 3));  % Seebeck coefficients in µV/°C
% === Calculate Seebeck Voltage ===
% We want an (N x M) matrix: each row = material, each column = temperature
V_output = S_uV .* T;  % Automatic broadcasting (MATLAB R2016b+)
% Expand S_uV to a column vector and T to a row vector (for compatibility)
S_uV_col = S_uV(:);        % N x 1
T_row = T(:)';             % 1 x M
V_output = S_uV_col * T_row;  % N x M (material x temperature)
% === Plot Seebeck Voltage vs Temperature ===
figure; hold on;
for i = 1:numMaterials
    plot(T, V_output(i, :), 'LineWidth', 2, ...
        'DisplayName', sprintf('%s (%s)', materialNames{i}, materialTypes{i}));
end
xlabel('Temperature (°C)');
ylabel('Seebeck Voltage (µV)');
title('Seebeck Voltage vs Temperature for Thermocouple Materials');
legend('show', 'Location', 'northwest');
```

```
grid on;
% === Print Table of Coefficients ===
fprintf('\n%-20s %-10s %-15s\n', 'Material', 'Type', 'Seebeck (µV/°C)');
fprintf('%s\n', repmat('-', 1, 50));
for i = 1:numMaterials
    fprintf('%-20s %-10s %-15.2f\n', materialNames{i}, materialTypes{i}, S_uV(i));
end
clc; clear;
% --- Geometric and Material Properties ---
L = 200e-6;         % Beam length (m)
b = 50e-6;          % Beam width (m)
h_sub = 2e-6;       % Substrate thickness (m)
h_piezo = 1e-6;     % Piezo layer thickness (m)
E_sub = 160e9;      % Young's modulus of substrate (Pa)
d31 = -190e-12;     % Piezoelectric strain coefficient (m/V)
% --- Electrical Excitation ---
V = 0:5:100;        % Voltage range (V)
% --- Bending Induced Displacement (Analytical Approximation) ---
% Tip deflection due to piezo strain: Δ = (3 * d31 * V * L^2) / (2 * h_sub)
delta_tip = (3 * abs(d31) * V * L^2) ./ (2 * h_sub);
% --- Plotting Results ---
figure;
plot(V, delta_tip * 1e6, 'r-', 'LineWidth', 2);  % Convert to µm
xlabel('Applied Voltage (V)');
ylabel('Tip Displacement (µm)');
title('Piezoelectric Cantilever Actuator: Tip Displacement vs Voltage');
grid on;
```

OUTPUT WAVEFORM:



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PROCEDURE:

1. Define Geometry and Material Properties.
2. Calculate Structural Parameters.
3. Define External Excitation Function.
4. Define the System Dynamics (ODE Function).
5. Calculate Generated Voltage.

INFERENCE:

RESULT:

Date: Expt. No. : 3	**Design and simulation of comb drive actuators**

AIM:

To design and simulation of thermocouples for various materials.

SOFTWARE REQUIRED:

- MATLAB
- Windows 11 OS

THEORY:

A comb drive actuator is a type of electrostatic micro-actuator commonly used in Micro-Electro-Mechanical Systems (MEMS). It consists of interdigitated comb-shaped electrodes—one set fixed and the other movable. When a voltage is applied, electrostatic forces pull the movable comb fingers towards the fixed ones, causing in-plane linear motion.

Comb-drives are microelectromechanical actuators, often used as linear actuators, which utilize electrostatic forces that act between two electrically conductive combs. Comb drive actuators typically operate at the micro- or nanometer scale and are generally manufactured by bulk micromachining or surface micromachining a silicon wafer substrate.

The attractive electrostatic forces are created when a voltage is applied between the static and moving combs causing them to be drawn together. The force developed by the actuator is proportional to the change in capacitance between the two combs, increasing with driving voltage, the number of comb teeth, and the gap between the teeth. The combs are arranged so that they never touch (because then there would be no voltage difference). Typically the teeth are arranged so that they can slide past one another until each tooth occupies the slot in the opposite comb.

Restoring springs, levers, and crankshafts can be added if the motor's linear operation is to be converted to rotation or other motions.

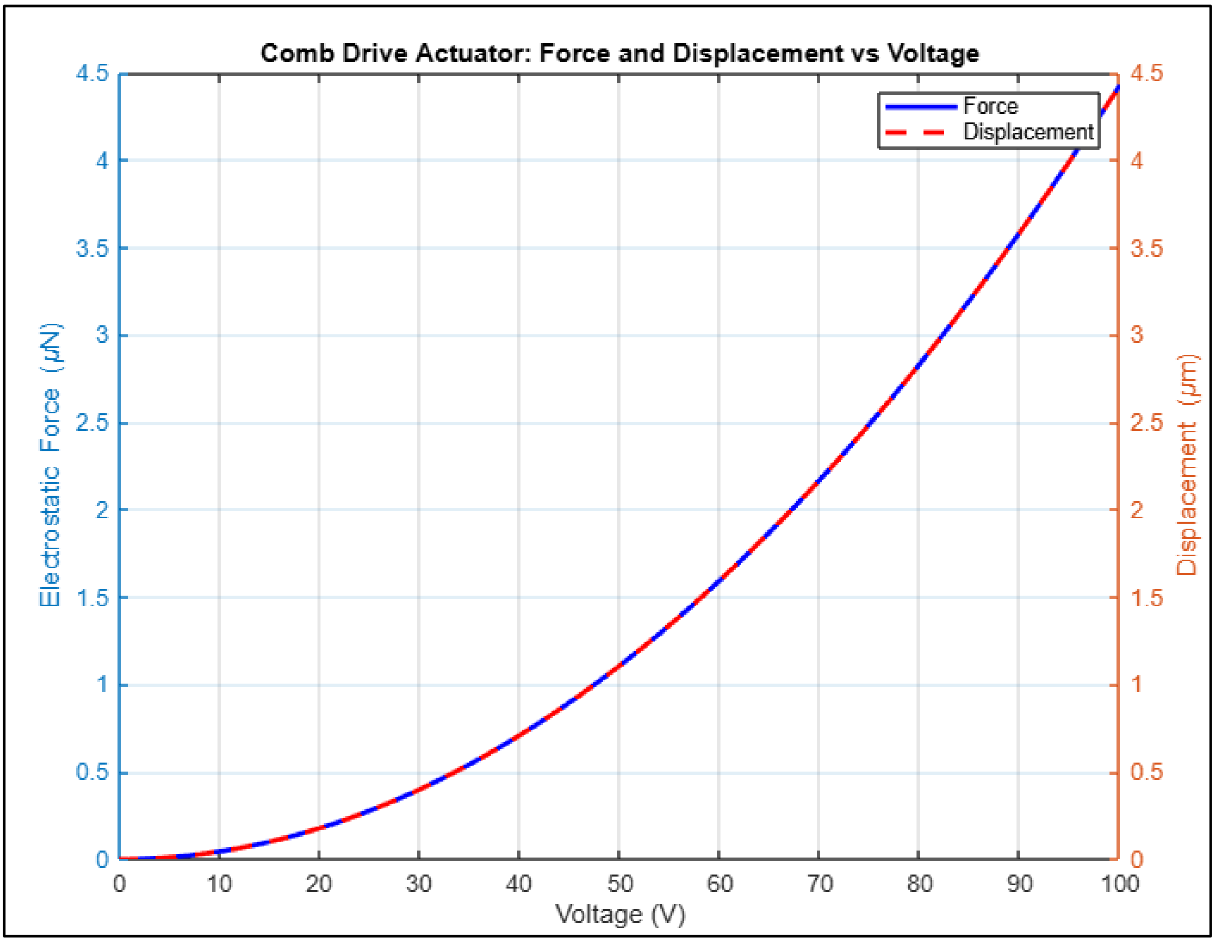
Electrostatic Force Equation:

$$F = \frac{1}{2} \cdot \frac{dC}{dx} \cdot V^2$$

PROGRAM:

```
clc; clear;

% Constants

epsilon0 = 8.854e-12; % Permittivity of free space (F/m)

% Design parameters

N = 100;            % Number of finger pairs

t = 2e-6;           % Finger thickness (m)

g = 2e-6;           % Gap between fingers (m)

k = 1;              % Spring constant (N/m)

% Voltage range

V = 0:1:100;        % Applied voltage (V)

% Electrostatic force calculation

F = 0.5 * N * epsilon0 * (t / g) * V.^2;  % Force in Newtons

% Displacement using Hooke's law: x = F/k

x = F / k;          % Displacement in meters

% Plotting

figure;

yyaxis left

plot(V, F * 1e6, 'b-', 'LineWidth', 2); % Force in microNewtons

ylabel('Electrostatic Force (\muN)');

yyaxis right

plot(V, x * 1e6, 'r--', 'LineWidth', 2); % Displacement in micrometers

ylabel('Displacement (\mum)');

xlabel('Voltage (V)');

title('Comb Drive Actuator: Force and Displacement vs Voltage');

legend('Force', 'Displacement');

grid on;
```

OUTPUT WAVEFORM:



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PROCEDURE:

1. Define Application Requirements
2. Choose Design Parameters
3. Calculate Required Force and Voltage
4. Spring Design
5. Simulation MATLAB

INFERENCE:

RESULT: